

# EMILY QI WANG

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## Education

### Olin College of Engineering

- Candidate for B.S. in Electrical and Computer Engineering
- Clare Boothe Luce Research Scholar (see Experience: Project Eye-Helper)

**Class of 2016**

Current GPA: 3.73

### University of Edinburgh (Study Abroad)

- Visiting third-year student in the School of Informatics

**Spring 2015**

## Experience

### Olin College Crowdsourcing and Machine Learning Lab

*Project Eye-Helper: Creating Assistive Technologies for Blind Grocery Shoppers*

*Spring 2014-Present*

- Designing a multi-device system to help the Blind shop independently and efficiently
- Spearheading the application development and feature integration between various platforms: Google Glass, Android smartphones, and node.js
- Implementing computer vision algorithms to track desired grocery items as the user moves throughout the store (combination of SIFT and meanshift techniques)
- Codesigning with members of the local blind and visually impaired community to ensure the product is user-friendly and will have a positive impact on the lives of users

### Olin College Curriculum

*Computer Architecture Course Assistant*

*Fall 2014*

- Working with electrical and computer engineering students as they build CPUs from the ground up.
- Includes modeling digital logic structures with Verilog, understanding the MIPS architecture, and project-based learning with FPGAs.

*Design Nature Course Assistant*

*Fall 2013*

- Mentoring first-year students in their first encounters with bioinspired design and rapid prototyping (designing for specific users, concept iterations, CAD in SolidWorks, fabrication)

### Relevant Coursework

*A Computational Introduction to Robotics*

*Fall 2014*

- Project-based approach to learning the algorithmic/software engineering side of robotic systems
- Topics include mastery of ROS, mobile robotic platforms, SLAM, computer vision, and navigation

*Computer Graphics - Cross-Registration at Wellesley College (CS-307)*

*Fall 2014*

- Program 3D graphics models using the THREE.js library and its WebGL capabilities
- Gain expertise with mathematical representations + coordinate systems for graphics systems and techniques for realistic rendering of objects, lighting, and animation

*User-Oriented Collaborative Design*

*Spring 2014*

- Collaborated with team of four to design an impactful product for volunteer physicians
- Final deliverable: The S2, an add-on intelligent note taker that will fill out electronic medical records, allowing for maximum doctor-patient interaction without medical records getting in the way

*Computer Architecture Final Project - Binary Clock*

*Fall 2013*

- Built a clock that displays everyday time in binary numbers with Verilog scripts running on a Spartan-3 FPGA and colorful LEDs
- Code is open source and repository is accessible on Github (greenteawarrior/BinaryClock)

## Activities

*Co-founder of Computing Conversations*

*Fall 2014*

- Student-run club for weekly discussions about interesting topics in computer science and monthly code jams

*Machine Learning Co-Curricular Member*

*Spring 2014 - present*

- Student-driven group which meets weekly to explore and apply concepts in machine learning
- Activities include: teaching fellow members about machine learning techniques, kaggle competitions, reading papers about innovations in the field, and working on group projects (creating a robotic tour guide for Olin)

*Flutist in Olin Conductorless Orchestra*

*Fall 2012 - Spring 2014*

## Skills

- Computer/Electrical: Image Processing (OpenCV), Web Development (node.js, jQuery, HTML/CSS), Linux, Robotics (ROS Python), Version Control (GitHub), Mobile Development (Android), Arduino prototyping, Soldering, CAD (SolidWorks), LaTeX
- Foreign Languages : Japanese (proficient), Chinese (conversational)